Meven Lennon-Bertrand

Post-doctoral researcher – Picube team, Inria & IRIF, Université Paris Cité

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Academic Positions

Inria Starting Research Position
Équipe projet Picube
Sep. 2025—... | IRIF, Université Paris Cité

Research Associate
Principal Investigator: Neel Krishnaswami
Nov. 2022—Aug. 2025 | University of Cambridge

PhD Student
Supervised by Nicolas Tabareau
Sep. 2019—Sep. 2022 | Gallinette team, Université de Nantes
Research intern Feb.—Jul. 2019.

Main Research Contributions

Journal Articles

- Sozeau, Forster et al., *Correct and Complete Type Checking and Certified Erasure for Coq, in Coq* (JACM 2024)
- Lennon-Bertrand, Maillard et al., *Gradualizing the Calculus of Inductive Constructions* (TOPLAS 2022)

Conference Articles

- Adjedj, Lennon-Bertrand et al., *AdapTT: Functoriality for Dependent Type Casts* (POPL 2026-01, to appear)
- Lennon-Bertrand, *What Does It Take to Certify a Conversion Checker?* (FSCD 2025, best paper by junior researcher)
- Laurent, Lennon-Bertrand et al., *Definitional Functoriality for Dependent (Sub)Types* (ESOP 2024, distinguished artifact)
- Adjedj, Lennon-Bertrand et al., *Martin-Löf à la Coq* (CPP 2024, distinguished paper)
- Maillard, Lennon-Bertrand et al., *A Reasonably Gradual Type Theory* (ICFP 2022)
- Lennon-Bertrand, *Complete Bidirectional Typing for the Calculus of Inductive Constructions* (ITP 2021)

Formalisations

- MetaCoq (contributor)
- LogRel-Coq (leader)
- Coq-PartialFun (contributor)
- AutoSubst 2 (contributor)

Interests

Logic · Dependent types · Proof assistants · Typing Algorithms · Formalised mathematics

Education

Master 2 (Computer Science)
2018—2019 | ENS de Lyon

Master 2 (Mathematics)
Preparation to the Agrégation, received 10th
2017—2018 | ENS de Lyon

Master 1 (Mathematical Foundations of Computer Science)
2016—2017 | Nijmegen, Erasmus exchange

Bachelors (Comp. Sci. & Mathematics)
Double Bachelor
2015—2016 | ENS de Lyon

Teaching and Outreach

Lecturer: Proof Assistants
2024 | University of Cambridge

Lecturer: Denotational Semantics
2023, 2024 | University of Cambridge

Science Popularisation: CHantiers Arts, Sciences et Technologies
2019–2022 | Lycée Michelet, Nantes

Teaching Assistant: Maths & CS
2019—2022 | Université de Nantes

Main Research Results

Topic

My research is at the intersection between mathematics, computer science and logic, in the field of proof assistants. These software tools aim at helping their users in writing, manipulating and checking mathematical proofs. They both help in getting a very high degree of confidence in the validity of the mechanized proof, but also give access to many software tools to create, analyse and maintain such mechanized proofs. The proof assistants I am particularly interested in are those based on type theory and the Curry-Howard correspondance, in particular the Coq proof assistant.

I try to make these tools both more expressive and safer, often by building on concepts and ideas coming for programming language theory.

Main Research Projects

Subtyping & Definitional Functoriality

[LLM24; Adj+01] — Théo Laurent, Kenji Maillard

We study the addition of extra equations to depedent type theory, corresponding to fuctoriality of type formers. Based on these, we present a form of structural subtyping well-suited for dependent types, and the equivalence of its implicit and explicit presentations.

Formalised logical relations in Coq

[Adj+24; Len25] — A. Adjedj, K. Maillard, L. Pujet, P.-M. Pédrot

We formalised, in Coq, a proof of normalisation based on logical relations, improving a previous proof in Agda [AÖV17] on multiple points: weaker meta-theory and richer object language, proof of decidability of typing using bidirectional typing, and executable type-checker.

Bidirectional Calculus of Inductive Constructions

[Len21; SLF22; Soz+24]

I gave a precise bidirectional presentation for the Calculus of Inductive Constructions, showed in Coq its equivalence with the standard presentation, and built on this to certify the correctness of MetaCoq's type-checker with respect to its specification. This work also led to the discovery and fix of a bug in the type-checker of Coq.

Gradualization of the Calculus of Inductive Constructions

[Len+22; Mai+22] — Kenji Maillard, Nicolas Tabareau, Éric Tanter

We built an extension of the Calculus of Inductive Constructions, the type system behind Coq, to feature a form of dynamic typing, following ideas from the gradual typing research area. This open the way to incorporate the flexibility offered by dynamic typing in proof assistants.

Software Development: Formalisation Projects

MetaCoq

2020-... — The MetaCoq team

This large collaborative project aims at formalizing Coq in Coq itself, and to allow manipulating Coq terms in Coq in order to develop certified meta-programming tools. I mainly contribute to the theoretical aspect, and currently my largest addition is the formalisation of my work on bidirectional typing, in order to prove the correctness of the type-checking algorithm implemented as part of the project.

This work allowed detecting and fixing a bug in the kernel of Coq. MetaCoq's certified type-checker serves as a basis for the certified compilers CertiCoq – from Coq to CompCert's Clight – and ConCert – to Elm, Rust and smart contract languages.

LogRel-Coq

2022-... A. Adjedj, K. Maillard, P.-M. Pédrot, L. Pujet

This project formalises, in Coq, proofs by logical relations of difficult meta-theoretic properties of dependent type systems, in particular normalisation. It is complementary to MetaCoq, in the sense that the latter concentrates on obtaining a certified type-checker for a system that is as close as possible to "real" Coq, but admits normalisation. Instead, LogRel-Coq currently works on a simpler language, but avoids any form of axiom.

Coq-PartialFun

2023-... Théo Winterhalter, Kenji Maillard

Support library for the definition of non-structurally recursive functions.

AutoSubst 2

2023-... Adrian Dapprich, Yannick Forster, Kathrin Stark

Code generator dedicated to syntax with binders.

Research Activity

Supervised students

Arthur Adjedj – Subtyping in Dependent Type Theory

ENS Paris-Saclay long research internship

Oct. 2024 – Aug. 2025

Robin Jourde – Understanding the η Law for Functions in CIC

Master 2 internship

Jan. – Jul. 2023 Co-supervised with Nicolas Tabareau

Matthew Sirman – A Normalisation by Evaluation Implementation of a Type Theory with Observational Equality

Undergraduate final dissertation (4th year)

Nov. 2022 – May 2023 Co-supervised with Neel Krishnaswami

Matthew was distinguished as the best student in Part III CS, in part for his dissertation.

Research Visits & Invited Seminars

CASH Team Seminar

Oct. 17 2025 ENS de Lyon

Épicure Team Seminar

Dec. 18 2024 Irisa, Université de Rennes

Formalisation of Mathematics with Interactive Theorem Provers

Nov. 7 2024 Faculty of Mathematics, Cambridge

Big Specification Workshop

17 Oct. 2024 Newton Institute, Cambridge

ReCiProg & Gard Condition Day

04 Jun. 2024 Université de Nantes

A one-day workshop, organised by the ReCiProg ANR grant.

OASIS Seminar
10 May 2024 — University of Oxford

Kathrin Stark & Dependable Systems Group
4 – 8 Mar. 2024 — Heriot-Watt University

Deducteam Seminar
14 Dec. 2023 — Université Paris-Saclay

Proofs and Algorithms Seminar
12 Dec. 2023 — LIX, École Polytechnique

PPS Seminar & Formath Seminar
7 & 11 Dec. 2023 — PPS, Université Paris Cité

Meta-programming, Quoting, and Modalities in Type Theory Workshop
16 – 20 Oct. 2023 — Université de Nantes

Conor McBride & Mathematically Structured Programming Group
26 – 30 Jun. 2023 — University of Strathclyde

CHoCoLa Seminar
Jan. 2023 — ENS de Lyon

LoVE Team Seminar
Dec. 2022 — Université Sorbonne Paris Nord

Andrej Bauer & Faculty of Mathematics and Physics Foundations Seminar
9 – 13 May 2022 — University of Ljubljana

Academic and Community Service

Types Steering Committee
2025-2028

In charge of the long-term organisation and supervision of the Types conference. I was elected a member during the general assembly of the 2025 edition, and will serve for 3 years.

SANDWICH Seminar (organiser)
2024-2025 — University of Cambridge

Internal seminar of the CLASH group.

Program Committee
JFLA 2026 · ITP 2025 · Types 2024

Artefact Evaluation Committee
ICFP 2022

Subreviewer
CPP 2026 · POPL 2026 · LICS 2025 · FSCD 2025 · CPP 2025 · CSL 2025

Proof Assistants Stack Exchange

2022-...

This website aims to answer questions around proof assistants in a community-based manner. I am in the top 5 most reputable users, and a moderator.

Elected Student Representative

2017-2018 ENS de Lyon

Publications

Most of my publications use the first author position to highlight the main contributor(s) to a project, either the main technical investigator(s) [Mai+22; Len+22; SLF22; LK23; LLM24; Adj+01], or the project lead [Soz+24]. In my field, conferences are the primary mean of publications, and they are typically as or even more selective than journals.

I support the Theoretical Computer Scientists for Future initiative. The possibility to travel plane-free is thus one of my main criteria when choosing a conference to publish in.

Journal

[Soz+24] 2024 Correct and Complete Type Checking and Certified Erasure for Coq, in Coq
Matthieu Sozeau, Yannick Forster, Meven Lennon-Bertrand, Jakob Botsch Nielsen, Nicolas Tabareau and Théo Winterhalter. *Journal of the ACM*.

[Len+22] 2022 Gradualizing the Calculus of Inductive Constructions
Meven Lennon-Bertrand, Kenji Maillard, Nicolas Tabareau and Éric Tanter. *ACM Transactions on Programming Languages and Systems*.

Conference

[Adj+01] 2026-01 AdapTT: Functoriality for Dependent Type Casts
Arthur Adjedj, Meven Lennon-Bertrand, Thibaut Benjamin and Kenji Maillard. *Proceeding of the ACM on Programming Languages* (to appear).

[Len25] 2025 What Does It Take to Certify a Conversion Checker?
Meven Lennon-Bertrand (best paper by junior researcher).

[LLM24] 2024 Definitional Functoriality for Dependent (Sub)Types
Théo Laurent, Meven Lennon-Bertrand and Kenji Maillard. *33rd European Symposium on Programming, ESOP 2024* (distinguished artifact).

[Adj+24] 2024 Martin-Löf *à la* Coq
Arthur Adjedj, Meven Lennon-Bertrand, Kenji Maillard, Pierre-Marie Pédrot and Loïc Pujet. *Proceedings of the 13th ACM SIGPLAN International Conference on Certified Programs and Proofs* (distinguished paper).

[Mai+22] 2022 A Reasonably Gradual Type Theory
Kenji Maillard, Meven Lennon-Bertrand, Nicolas Tabareau and Éric Tanter. *International Conference on Functional Programming*.

[Len21] 2021 Complete Bidirectional Typing for the Calculus of Inductive Constructions
Meven Lennon-Bertrand. *12th International Conference on Interactive Theorem Proving*.

Thesis

[Len22] 2022 Bidirectional Typing for the Calculus of Inductive Constructions
Meven Lennon-Bertrand.

Peer-reviewed workshops

[SLK24] **Implementing Observational Equality Using Normalisation by Evaluation**
2024 Matthew Sirman, Meven Lennon-Bertrand and Neel Krishnaswami. *30th International Conference on Types for Proofs and Programs*.

[Len24] **Towards a certified proof assistant kernel**
2024 Meven Lennon-Bertrand. *Meeting of the Working Group 6 of the European Research Network on Formal Proofs* (invited talk).

[LK23] **Decidable Type-Checking for Bidirectional Martin-Löf Type Theory**
2023 Meven Lennon-Bertrand and Neel Krishnaswami. *29th International Conference on Types for Proofs and Programs*.

[Mai+23] **Engineering logical relations for MLTT in Coq**
2023 Kenji Maillard, Arthur Adjedj, Meven Lennon-Bertrand and Loïc Pujet. *29th International Conference on Types for Proofs and Programs*.

[Len22a] **Equivalence between Typed and Untyped Algorithmic Conversions**
2022 Meven Lennon-Bertrand. *28th International Conference on Types for Proofs and Programs*.

[Len22b] **À bas l'η – Coq's troublesome η-conversion**
2022 Meven Lennon-Bertrand. *1st Workshop on the Implementation of Type Systems*.

[SLF22] **The Curious Case of Case: Correct & Efficient Representation of Case Analysis in Coq and MetaCoq**
2022 Matthieu Sozeau, Meven Lennon-Bertrand and Yannick Forster. *1st Workshop on the Implementation of Type Systems*.

Publication in open archives

[BR18] **Coalgebraic Determinization of Alternating Automata**
2018 Meven Bertrand and Jurriaan Rot. *arXiv*, DOI: 10.48550/ARXIV.1804.02546

Teaching and Science Outreach

Co-Lecturer: Denotational Semantics

Fall 2024 — University of Cambridge

With Thomas Bauereiss, we co-lecture a course on proof assistants for 4th year students. I was in charge of the Coq half of the course.

Lecturer: Denotational Semantics

Fall 2023, Fall 2024 — University of Cambridge

For two years in a row, I lectured the Denotational Semantics course for 3rd year students.

Teaching Assistant (64 hrs/year)

Sep. 2019—Jun. 2022 — Université de Nantes

During my PhD, I also worked as a teaching assistant. I taught various levels (1st to 3rd Bachelor years), themes (mathematics, applied and fundamental computer science), formats (lectures, exercise and computer sessions), and publics (specialists and non-specialists).

CHantiers Arts, Sciences et Technologies

2019–2022 — Théâtre Athénor & Lycée Professionnel Michelet, Nantes

Together with math researcher Bertrand Michel and theatre authors Rémi Checchetto and Sylvain Renard, we collaborated with a vocational high-school teachers to build workshops for their students around the theme of "Artificial Intelligence", in a broad sense. I implemented activities directly inspired by the *Computer Science Unplugged* project, and designed some of my own. These workshops culminated in an exhibition, created by the students.

Séminaire de la Détente Mathématique

2018—2019 Maison des Mathématiques et de l'Informatique, Lyon

A weekly seminar, aimed at being "relaxed" and accessible to both students and faculty, with talks often on unusual or fun topics. Many students would give their first seminar talk there. I organised the seminar with a team of students, and spoke there.

Category Theory Course

Sept 2018 – Jan 2019 ENS de Lyon

During my Master 2, I co-organized with a fellow student a category theory course for students of the ENS de Lyon. Although it was not integrated to the official curriculum, we kept it as close as possible to a proper lecture: it lasted for a semester of a 2-hour lecture per week, with a few dozen students, and we covered a large part of Awodey's *Category theory*.

Thesis

Defended at Université de Nantes, on June 24, 2022. Prepared in the Inria team Gallinette, affiliated to the Laboratoire des Sciences du Numérique de Nantes.

Chair of the Jury
Christine Paulin-Mohring (Professeure des Universités, Université Paris Sud)

PhD Advisor
Nicolas Tabareau (Directeur de Recherche, Inria Rennes)

Rapporteurs
Neel Krishnaswami (Associate Professor, University of Cambridge)
Conor McBride (Reader, University of Strathclyde)

Examiners
Jesper Cockx (Assistant Professor, TU Delft)
Herman Geuvers (Professor, Radboud University Nijmegen)
Hugo Herbelin (Directeur de Recherche, Inria Paris)
Assia Mahboubi (Directrice de Recherche, Inria Rennes)

Invited Member
Matthieu Sozeau (Chargé de Recherche, Inria Rennes)